Math 15 — Unit 7: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

Part A — Linear vs. Non-Linear, and Tables of Values

1. Each equation below represents a relation. Without graphing, sort the letters into the LINEAR row or the NON-LINEAR row.

(a) $y = 5x + 2$ (b) $y = x^2 - 4$ (c) $y = -4x$ (d) $y = 7$ (e) $y = \frac{1}{x}$ (f) $y = x^3 - 1$ (g) $y = -x - 9$ (h) $y = 4^x$

LINEAR: _____

NON-LINEAR: _____

2. Explain in your own words why $y = x^2 - 3$ is **not** a linear equation, but $y = -3x - 3$ **is** a linear equation.

3. Build a table of values for $y = 3x - 2$, using $x = -2, -1, 0, 1, 2$.

x	-2	-1	0	1	2
y					

4. Build a table of values for $y = -x + 5$, using $x = -2, -1, 0, 1, 2$.

x	-2	-1	0	1	2
y					

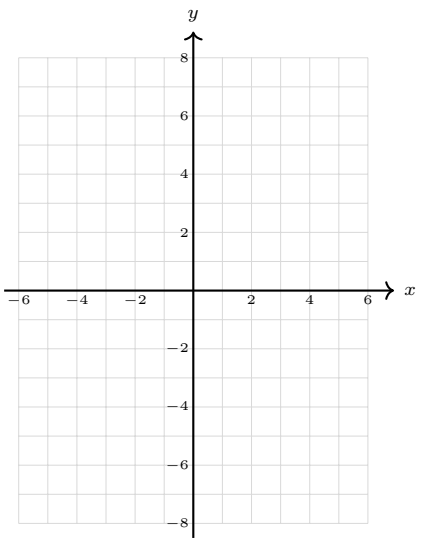
5. Build a table of values for $y = 2x + 1$, using $x = -2, -1, 0, 1, 2$.

x	-2	-1	0	1	2
y					

Part B — Tables of Values and Graphs

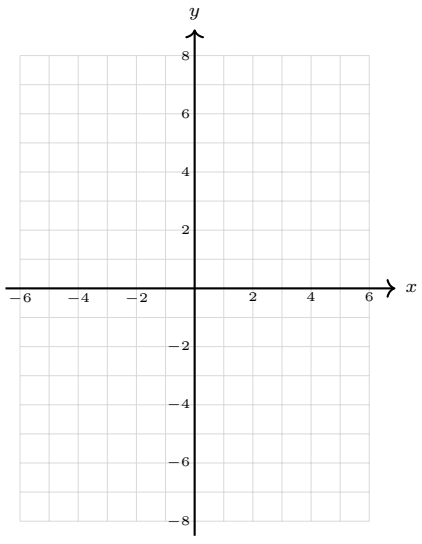
6. Build a table of values for $y = x + 2$, using $x = -2, -1, 0, 1, 2$. Then plot the points and graph the line.

x	-2	-1	0	1	2
y					



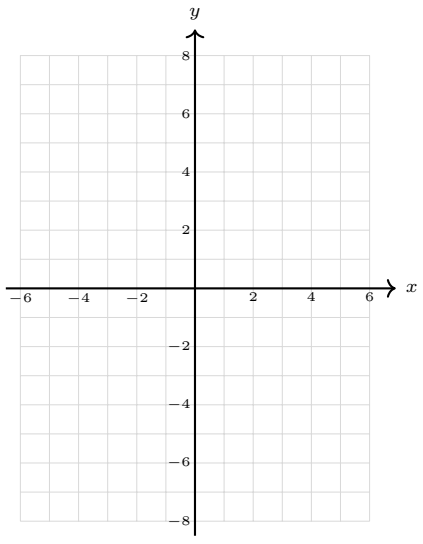
7. Build a table of values for $y = -3x + 2$, using $x = -2, -1, 0, 1, 2$. Then plot the points and graph the line.

x	-2	-1	0	1	2
y					



8. Build a table of values for $y = 2x - 4$, using $x = -2, -1, 0, 1, 2$. Then plot the points and graph the line.

x	-2	-1	0	1	2
y					



Part C — Does the Point Lie on the Line?

9. Does the point $(2, 7)$ lie on the line $y = 3x + 1$? Show your substitution.

10. Does the point $(4, -3)$ lie on the line $y = -x + 2$? Show your substitution.

11. Does the point $(-3, -4)$ lie on the line $y = 2x + 2$? Show your substitution.

12. Consider the equation $y = -2x + 6$. Determine whether each point lies on the line: $(1, 4)$, $(0, 5)$, $(-2, 10)$.

13. A table of values for a linear equation contains the points $(-1, -1)$, $(0, 2)$, $(1, 5)$, and $(2, 8)$. Without writing the equation, use the pattern in the y-values to predict the y-value when $x = 3$. Explain your reasoning.

Part D — Multiple Choice and Numerical Response

14. *(Multiple Choice)* Which of the following points lies on the line $y = 4x - 5$?
A. $(0, -5)$ B. $(0, 5)$ C. $(5, 0)$ D. $(-5, 0)$

15. *(Numerical Response)* The point $(k, 20)$ lies on the line $y = 3x + 2$. The value of k is _____.

Answer Key. 1. LINEAR: a, c, d, g; NON-LINEAR: b, e, f, h. **2.** $y = x^2 - 3$ has x squared, so its graph is a parabola, not a line; $y = -3x - 3$ is in the form $y = mx + b$ with x to the first power only. **3.** $y : -8, -5, -2, 1, 4$. **4.** $y : 7, 6, 5, 4, 3$. **5.** $y : -3, -1, 1, 3, 5$. **6.** $y : 0, 1, 2, 3, 4$; straight line through these points, extended across the grid. **7.** $y : 8, 5, 2, -1, -4$; straight line through these points, extended across the grid. **8.** $y : -8, -6, -4, -2, 0$; straight line through these points, extended across the grid. **9.** Yes: $3(2) + 1 = 7$. **10.** No: $-(4) + 2 = -2 \neq -3$. **11.** Yes: $2(-3) + 2 = -4$. **12.** $(1, 4)$ yes; $(0, 5)$ no; $(-2, 10)$ yes. **13.** y rises by 3 for each increase of 1 in x, so at $x = 3$, $y = 8 + 3 = 11$. **14.** A. **15.** $k = 6$.

Lesson 2 Assignment *Show your work.*

Part A — Reading m and b Directly

1–6. State the slope m and the y-intercept b of each equation.

1. $y = 6x + 5$
2. $y = -2x + 9$
3. $y = x - 4$
4. $y = -x + 1$
5. $y = 8x$
6. $y = -3$

Part B — Rearranging Into $y = mx + b$

7–12. Rearrange each equation into slope–intercept form, then state m and b.

7. $x + y = 10$
8. $4x + y = 3$
9. $-6x + y = -2$
10. $2y = 10x + 6$
11. $2x + 2y = 16$
12. $8x + 4y = 28$

Part C — Tables, Applications and Reasoning

13. Does the table below match $y = 2x + 1$? Check each ordered pair and explain.

x	-1	0	1	2
y	-1	1	3	5

14. Does the table below match $y = 5x - 3$? Check each ordered pair and explain.

x	0	1	2	3
y	-3	2	7	11

15. A moving company charges cost $= 50 + 30h$, where h is the number of hours worked and cost is in dollars.
(a) Identify the starting cost (b) and explain what it represents.
(b) Identify the cost per hour (m) and explain what it represents.
(c) Determine the total cost of a 6-hour job.

16. In your own words, explain why you must isolate y before you can read the slope and y-intercept directly from an equation like $3x + y = 12$.

Answer Key. 1. $m = 6$, $b = 5$. **2.** $m = -2$, $b = 9$. **3.** $m = 1$, $b = -4$. **4.** $m = -1$, $b = 1$. **5.** $m = 8$, $b = 0$. **6.** $m = 0$, $b = -3$. **7.** $y = -x + 10$; $m = -1$, $b = 10$. **8.** $y = -4x + 3$; $m = -4$, $b = 3$. **9.** $y = 6x - 2$; $m = 6$, $b = -2$. **10.** $y = 5x + 3$; $m = 5$, $b = 3$. **11.** $y = -x + 8$; $m = -1$, $b = 8$. **12.** $y = -2x + 7$; $m = -2$, $b = 7$. **13.** Yes — every pair satisfies $y = 2x + 1$ (e.g. $x = 2$ gives $y = 5$). **14.** No — at $x = 3$ the equation gives $y = 12$, but the table shows 11. **15.** (a) $b = 50$: the flat fee charged before any hours are worked. (b) $m = 30$: the extra cost per hour. (c) $50 + 30(6) = \$\mathbf{230}$. **16.** y is not alone — the $3x$ term is still attached, so the numbers as written are not m and b; rearranging to $y = -3x + 12$ shows the true slope -3 and y-intercept 12.

Lesson 3 Assignment *Show your work.*

Part A — Reading the Slope and the y-Intercept

1. State the slope m and the y-intercept b for each equation. (No graphing needed.)

(a) $y = 6x + 1$ (b) $y = -4x + 7$ (c) $y = \frac{1}{2}x - 3$ (d) $y = -\frac{3}{5}x$ (e) $y = 8 - 2x$

2. Write the y-intercept of each equation as an ordered pair $(0, b)$.

(a) $y = 3x - 9$ (b) $y = -7x + 2$ (c) $y = \frac{4}{5}x$ (d) $y = 10 - x$

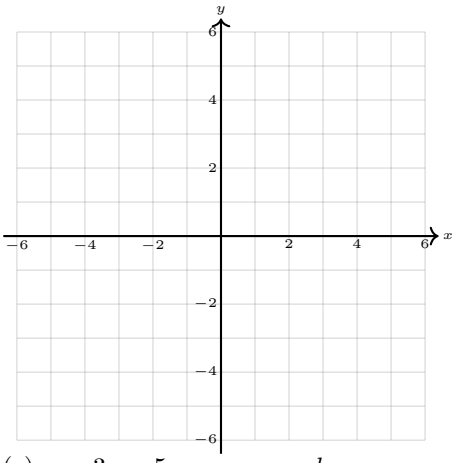
3. Write the slope of each equation as a fraction $\frac{\text{rise}}{\text{run}}$.

(a) $y = 4x - 1$ (b) $y = -6x + 3$ (c) $y = \frac{3}{7}x - 2$ (d) $y = -\frac{5}{2}x + 1$

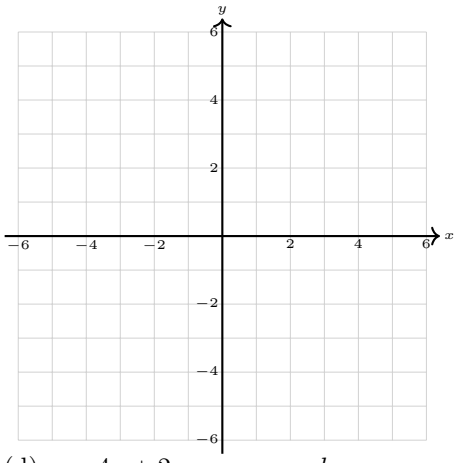
Part B — Graphing from $y = mx + b$

4. For each equation, state the slope and y-intercept, then graph the line by plotting the y-intercept, using the slope to find a second point, and drawing the line through both points.

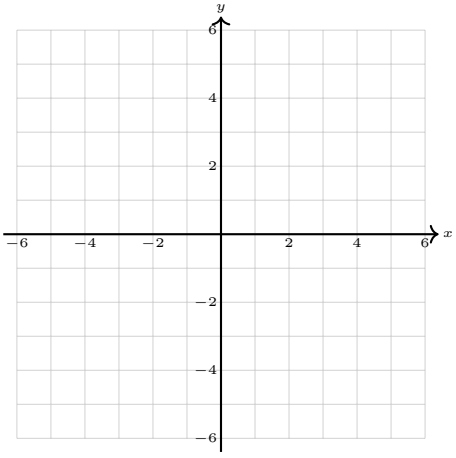
(a) $y = x + 1$ $m =$____ $b =$____



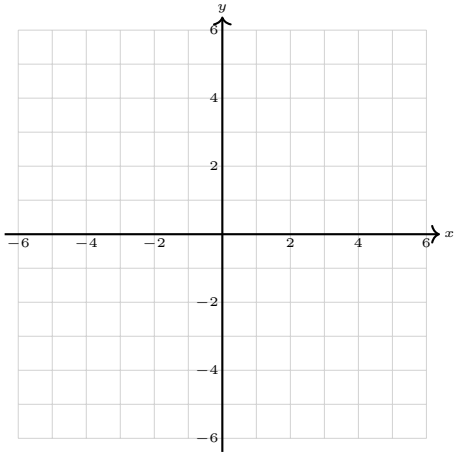
(b) $y = 2x - 2$ $m =$____ $b =$____



(c) $y = 3x - 5$ $m =$____ $b =$____

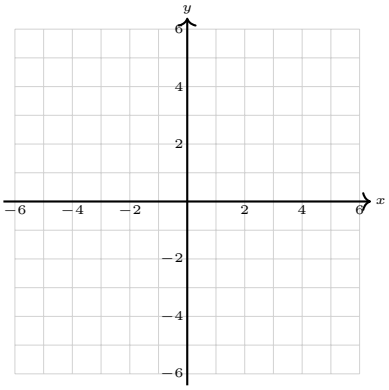


(d) $y = 4x + 2$ $m =$____ $b =$____

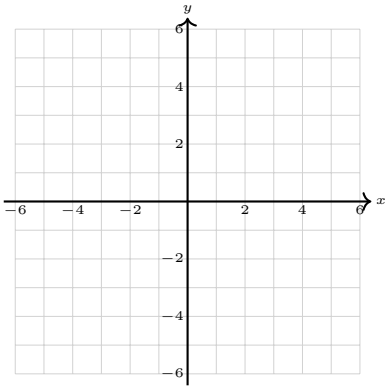


5. Same instruction as Question 4. Each of these lines has a **negative** slope.

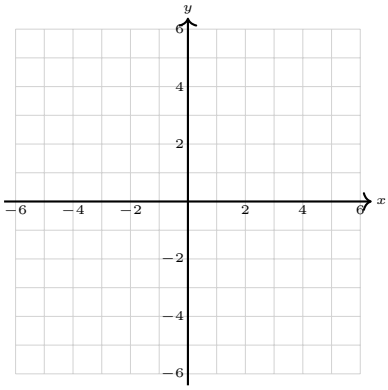
(a) $y = -x + 4$
$m =$___ $b =$___



(b) $y = -2x + 3$
$m =$___ $b =$___

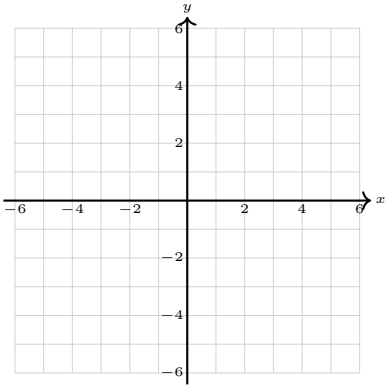


(c) $y = -3x - 1$
$m =$___ $b =$___

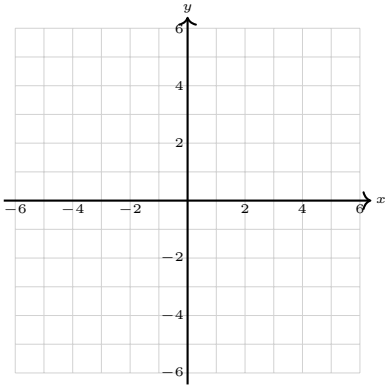


6. Same instruction as Question 4. Each of these lines has a **fractional** slope.

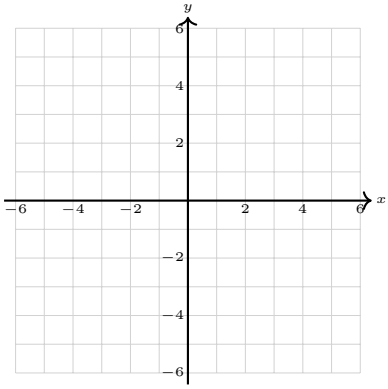
(a) $y = \frac{1}{2}x - 4$
$m =$___ $b =$___



(b) $y = \frac{3}{4}x + 1$
$m =$___ $b =$___



(c) $y = -\frac{2}{3}x + 2$
$m =$___ $b =$___



Part C — Short Answer and Application

7. Explain why you always start graphing at the y-intercept rather than at some other point on the line.

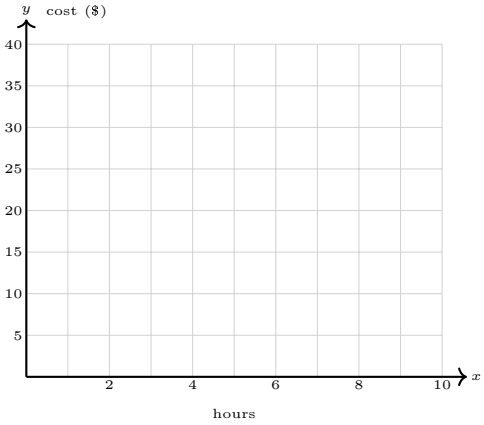
8. A line has slope -4. Write this slope as two different but equivalent fractions $\frac{\text{rise}}{\text{run}}$, and explain why both fractions produce the same line.

9. A bike rental company charges according to the equation $y = 2x + 15$, where x is the number of hours rented and y is the total cost in dollars.

(a) State the slope and y-intercept of this equation.

(b) What does the y-intercept represent in this situation?

(c) Graph the line using the slope and y-intercept method.



Part D — Multiple Choice and Numerical Response

10. *(Multiple Choice)* When graphing $y = -5x + 3$ using the slope-intercept method, which point should you plot *first*?
A. $(0, -5)$ B. $(0, 3)$ C. $(3, 0)$ D. $(-5, 0)$

11. *(Multiple Choice)* A line has equation $y = \frac{3}{4}x - 2$. Starting at the y-intercept, which move locates a second point on the line?
A. Right 4, up 3 B. Right 3, up 4
C. Right 4, down 3 D. Right 2, up $\frac{3}{4}$

12. *(Multiple Choice)* Which statement is *false* for the line $y = -2x + 5$?
A. The y-intercept is 5. B. The graph falls from left to right. C. The slope can be written as $\frac{-2}{1}$ or $\frac{2}{-1}$. D. The line passes through the point $(1, 7)$.

Use the following information to answer Questions 13 and 14. Consider the line with equation $y = -\frac{4}{3}x + 4$. The line intersects the x-axis at P and the y-axis at Q. Triangle POQ is formed, where O is the origin. (Graph the line first: P is where it crosses the x-axis.)

13. *(Numerical Response)* The area of $\triangle POQ$, in square units, is _____.

14. *(Numerical Response)* The perimeter of $\triangle POQ$, in units, is _____.

Answer Key. 1. (a) $m = 6, b = 1$ (b) $m = -4, b = 7$ (c) $m = \frac{1}{2}, b = -3$ (d) $m = -\frac{3}{5}, b = 0$ (e) $m = -2, b = 8$. **2.** (a) $(0, -9)$ (b) $(0, 2)$ (c) $(0, 0)$ (d) $(0, 10)$. **3.** (a) $\frac{4}{1}$ (b) $\frac{-6}{1}$ (c) $\frac{3}{7}$ (d) $\frac{-5}{2}$. **4.** (a) $(0, 1)$, slope 1 (b) $(0, -2)$, slope 2 (c) $(0, -5)$, slope 3 (d) $(0, 2)$, slope 4. **5.** (a) $(0, 4)$, slope -1 (b) $(0, 3)$, slope -2 (c) $(0, -1)$, slope -3 — all fall to the right. **6.** (a) $(0, -4)$, slope $\frac{1}{2}$ (b) $(0, 1)$, slope $\frac{3}{4}$ (c) $(0, 2)$, slope $-\frac{2}{3}$. **7.** $(0, b)$ is read straight from the equation with no calculation, so it is a guaranteed reliable starting point. **8.** $-4 = \frac{-4}{1}$ (down 4, right 1) $= \frac{4}{-1}$ (up 4, left 1); both land on the same line, approached in opposite directions. **9.** (a) $m = 2$, $b = 15$ (b) the flat \$15 starting fee, charged before any hours (c) line through $(0, 15)$, slope 2 (i.e. right 5 h, up \$10). **10.** B. **11.** A. **12.** D (at $x = 1$, $y = 3$). **13.** 6 ($P(3, 0)$, $Q(0, 4)$). **14.** 12 $(3 + 4 + 5)$.

Lesson 4 Assignment *Show your work.*

Part A — From a Given Slope and y-Intercept

1. Write the equation of each line.

(a) slope 3, y-int 5

(b) slope -4, y-int -2

(c) slope $\frac{1}{2}$, y-int 8

(d) slope $-\frac{2}{3}$, y-int -6

(e) slope 7, through $(0, 0)$

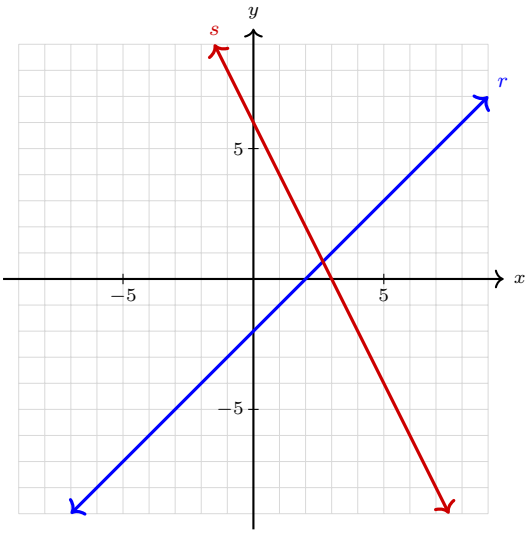
(f) y-int -10, slope 1

(g) slope -5, y-int 0

(h) slope $\frac{2}{5}$, through $(0, 11)$

Part B — From Graphs and Tables

2. Each line on the grid passes through points with integer coordinates. Determine the equation of each line.



Line r
Slope: _____ y-intercept: _____
Equation: _____

Line s
Slope: _____ y-intercept: _____
Equation: _____

3. Determine the equation of the line represented by each table.

(a)

x	0	1	2	3
y	6	9	12	15

(b)

x	0	1	2	3
y	20	16	12	8

4. A candle shop tracks the height of a candle as it burns.

Time, t (h)	0	1	2	3
Height, h (cm)	18	15	12	9

(a) State the slope and the y-intercept for this table.

(b) Write an equation for the height h in terms of the time t.

(c) In this situation, what does the slope represent? What does the y-intercept represent?

Part C — Applications

5. A gym charges a \$30 sign-up fee plus \$15 per month for a membership.
(a) Write an equation for the total cost C after t months.

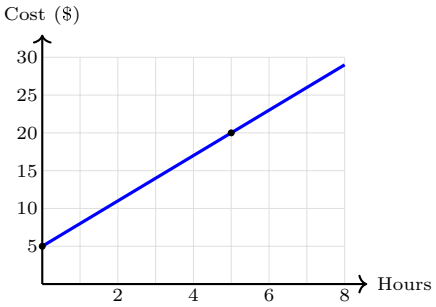
(b) Find the cost of a 6-month membership.

6. A candle is 24 cm tall and burns down at a constant rate of 3 cm per hour.
(a) Write an equation for the height h of the candle after t hours.

(b) Find the height of the candle after 5 hours.

7. A water tank already contains 15 L of water when a hose begins filling it at a constant rate of 4 L per minute.
(a) Write an equation for the volume V after t minutes.

(b) Find the volume after 10 minutes.



8. The graph at left shows the cost of renting a bicycle for a given number of hours. Each grid square on the vertical axis represents \$5. Determine the equation for the cost C in terms of the number of hours h.

9. The equation $y = 60x + 50$ models the total charge y (in dollars) for a plumber's house call that takes x hours. In this situation, what does the number 60 represent? What does the number 50 represent?

Part D — Multiple Choice and Numerical Response

10. *(Multiple Choice)* Which equation has slope -4 and y-intercept 9?
A. $y = 9x - 4$ B. $y = -4x + 9$ C. $y = 4x - 9$ D. $y = -9x + 4$

11. *(Multiple Choice)* A table shows the points $(0, 5)$, $(1, 9)$, $(2, 13)$, and $(3, 17)$. Which equation matches this table?
A. $y = 4x + 5$ B. $y = 5x + 4$ C. $y = 4x - 5$ D. $y = -4x + 5$

12. *(Multiple Choice)* A moving company charges a flat fee of \$50 plus \$20 per hour of labour. Which equation models the total cost C after h hours?
A. $C = 50h + 20$ B. $C = 20h + 50$ C. $C = 70h$ D. $C = 20h - 50$

13. *(Numerical Response)* A parking garage charges a \$4 entry fee plus \$2 per hour, so the cost equation is $y = 2x+4$. The total cost to park for 9 hours is _____ dollars.

14. *(Numerical Response)* A moving company charges a flat fee of \$80 plus \$25 per hour of labour. If the total cost equation is written in the form $C = mh+b$, the value of $m + b$ is _____.

15. *(Numerical Response)* A candle is 30 cm tall and burns down 3 cm every hour. Using $h = mt + b$ for its height after t hours, the candle's height after 6 hours, in cm, is _____.

Answer Key. 1. (a) $y = 3x + 5$ (b) $y = -4x - 2$ (c) $y = \frac{1}{2}x + 8$ (d) $y = -\frac{2}{3}x - 6$ (e) $y = 7x$ (f) $y = x - 10$ (g) $y = -5x$ (h) $y = \frac{2}{5}x + 11$. **2.** r: $y = x - 2$; s: $y = -2x + 6$. **3.** (a) $y = 3x + 6$ (b) $y = -4x + 20$. **4.** (a) $m = -3$, $b = 18$ (b) $h = -3t + 18$ (c) slope: height drops 3 cm every hour; y-intercept: the candle started at 18 cm before any burning. **5.** (a) $C = 15t + 30$ (b) \$120. **6.** (a) $h = -3t + 24$ (b) 9 cm. **7.** (a) $V = 4t + 15$ (b) 55 L. **8.** $C = 3h + 5$. **9.** 60: cost per hour of labour; 50: flat call-out fee charged even before any labour. **10.** B. **11.** A. **12.** B. **13.** 22. **14.** 105. **15.** 12 ($h = -3t + 30$).

Lesson 5 Assignment *Show your work.*

Part A — Substitute and Solve

1. Use the substitute-and-solve method to determine the equation of the line with the given slope through the given point.
(a) slope 2, point $(1, 8)$ (b) slope 5, point $(2, 13)$ (c) slope 3, point $(4, 17)$

(d) slope 1, point $(6, 10)$ (e) slope 7, point $(3, 25)$ (f) slope 4, point $(5, 23)$

2. Determine the equation of the line with the given slope through the given point.
(a) slope -2, point $(3, 1)$ (b) slope -5, point $(2, -3)$ (c) slope -3, point $(4, -7)$

(d) slope -1, point $(8, 2)$

3. Determine the equation of the line with the given slope through the given point. Each point has two negative coordinates.
(a) slope 2, point $(-3, -4)$ (b) slope -4, point $(-2, -9)$ (c) slope 3, point $(-5, -2)$

(d) slope -6, point $(-1, -1)$

4. Determine the equation of the line with the given slope through the given point.
(a) slope $\frac{1}{2}$, point $(6, 7)$ (b) slope $\frac{1}{3}$, point $(9, 8)$

(c) slope $-\frac{3}{4}$, point $(8, 1)$ (d) slope $-\frac{1}{2}$, point $(-4, 9)$

5. First decide whether the given point *is* the y-intercept. If it is, state b right away; if not, use the substitute-and-solve method.
(a) slope 4, point $(0, -6)$ (b) slope 4, point $(2, -6)$

6. Determine the equation of each line, then check it by substituting the given point back into your final equation.
(a) slope 5, point $(3, 19)$ (b) slope -2, point $(5, -3)$

Part B — Applications

7. A hiker descends a mountain trail at a constant rate. After 3 hours her elevation is 1200 m, and she descends at 150 m per hour. Let E be her elevation (in metres) after t hours.

(a) Write an equation for E in terms of t.

(b) What was her elevation at $t = 0$?

(c) Predict her elevation after 5 hours.

8. A gym's membership grows at a constant rate. After 4 months the gym has 210 members, and membership grows by 15 members every month. Let M be the number of members after t months.

(a) Write an equation for M in terms of t.

(b) How many members were there at $t = 0$?

(c) Predict the membership after 10 months.

9. A young tree grows at a constant rate. At age 12 years it is 47 ft tall, and it grows 3 ft every year. Let h be the height (in feet) at age w years.

(a) Write an equation for h in terms of w.

(b) How tall was the tree at age 0?

(c) Predict the height at age 20.

Part C — Multiple Choice and Numerical Response

10. *(MC)* Which is the equation of the line with slope 3 through $(2, 11)$?
A. $y = 3x - 5$ B. $y = 3x + 5$
C. $y = 3x + 11$ D. $y = 5x + 3$

11. *(MC)* A line has slope -4 and passes through $(3, -1)$. What is the value of b in $y = -4x + b$?
A. -11 B. 13 C. 11 D. -13

12. *(MC)* A line has slope 6 and passes through $(1, 2)$. Which is the correct equation in $y = mx + b$ form?
A. $y = 6x + 4$ B. $y = 6x - 2$
C. $y = -6x - 4$ D. $y = 6x - 4$

13. *(NR)* A line has slope 7 and passes through $(2, 19)$. The value of b in $y = 7x + b$ is _____.

14. *(NR)* A line has slope -3 and passes through $(-2, 10)$. Written as $y = -3x + b$, the value of b is _____.

15. *(NR)* A tree is 15 ft tall at age 6 and grows 2 ft per year; its height at age a is $h = 2a + b$. The value of b is _____.

Answer Key. **1.** (a) $y = 2x + 6$ (b) $y = 5x + 3$ (c) $y = 3x + 5$ (d) $y = x + 4$ (e) $y = 7x + 4$ (f) $y = 4x + 3$. **2.** (a) $y = -2x + 7$ (b) $y = -5x + 7$ (c) $y = -3x + 5$ (d) $y = -x + 10$. **3.** (a) $y = 2x + 2$ (b) $y = -4x - 17$ (c) $y = 3x + 13$ (d) $y = -6x - 7$. **4.** (a) $y = \frac{1}{2}x + 4$ (b) $y = \frac{1}{3}x + 5$ (c) $y = -\frac{3}{4}x + 7$ (d) $y = -\frac{1}{2}x + 7$. **5.** (a) is the y-int., $b = -6$: $y = 4x - 6$ (b) not the y-int., $b = -14$: $y = 4x - 14$. **6.** (a) $y = 5x + 4$ (b) $y = -2x + 7$. **7.** (a) $E = -150t + 1650$ (b) 1650 m (c) 900 m. **8.** (a) $M = 15t + 150$ (b) 150 (c) 300. **9.** (a) $h = 3w + 11$ (b) 11 ft (c) 71 ft. **10.** B. **11.** C. **12.** D. **13.** 5. **14.** 4. **15.** 3.

Lesson 6 Assignment *Show your work.*

Part A — Slope First, Then the Equation

1. Find the slope of the line through each pair of points.

(a) $(1, 2)$ and $(4, 11)$

(b) $(0, 5)$ and $(3, -1)$

(c) $(-2, -3)$ and $(2, 5)$

(d) $(-5, 7)$ and $(0, -3)$

2. Determine the equation, in the form $y = mx + b$, of the line through each pair of points.

(a) $(1, 4)$ and $(3, 10)$

(b) $(2, 1)$ and $(5, 13)$

(c) $(0, -2)$ and $(4, 10)$

(d) $(-1, 6)$ and $(2, -3)$

(e) $(-3, -8)$ and $(3, 4)$

(f) $(-4, 9)$ and $(4, -7)$

3. Determine the equation, in the form $y = mx + b$, of the line through each pair of points. Then check it by substituting the second point into your equation.

(a) $(2, 3)$ and $(5, 15)$

(b) $(-3, 10)$ and $(1, -2)$

(c) $(-6, -4)$ and $(0, 8)$

4. Determine the equation, in the form $y = mx + b$, of the line through each pair of points. The slopes here are fractions — keep them exact.

(a) $(-6, 1)$ and $(2, 5)$

(b) $(0, 7)$ and $(6, 3)$

(c) $(-5, -9)$ and $(5, 3)$

(d) $(-4, 5)$ and $(2, 2)$

Part B — Special Cases

5. Determine the equation, in the form $y = mx + b$, of the line through each pair of points. *Hint:* one point in each pair already sits on the y-axis.

(a) $(0, -6)$ and $(3, 0)$

(b) $(0, 9)$ and $(4, 1)$

6. Determine the equation of the line through each pair of points.

(a) $(-3, 5)$ and $(6, 5)$

(b) Explain why the process from this lesson cannot produce an equation of the form $y = mx + b$ for the points $(7, 4)$ and $(7, -2)$. *Hint:* what happens to the slope?

Part C — Applications

7. A cyclist rides away from home at a constant speed. After 2 hours she is 15 km from home; after 5 hours she is 33 km from home.

(a) Determine the equation relating her distance d (km) from home and her riding time t (hours).

(b) What do the slope and the d-intercept represent?

(c) Predict her distance from home after 8 hours.

8. A phone charges at a constant rate. After 1 minute the battery reads 42%; after 6 minutes it reads 72%.

(a) Determine the equation relating the battery percentage p and the charging time t (minutes).

(b) Predict the battery percentage after 10 minutes of charging.

9. A young tree was 40 cm tall at age 2 years and 100 cm tall at age 5 years. Assuming constant growth, determine the equation relating its height h (cm) and its age a (years).

Part D — Multiple Choice, Numerical Response and Challenge

10. *(MC)* The equation of the line through $(3, 7)$ and $(6, 19)$ is

A. $y = 4x - 5$ B. $y = 4x + 5$
C. $y = 3x - 5$ D. $y = 4x - 12$

11. *(MC)* A line passes through $(-2, 8)$ and $(2, 0)$. Which is its equation?

A. $y = -2x + 4$ B. $y = 2x + 4$
C. $y = -2x - 4$ D. $y = -4x + 4$

12. *(NR)* The line through $(1, -3)$ and $(4, 9)$ has equation $y = mx + b$. The value of $m - b$ is _____.

13. *(NR)* A line passes through $(0, 5)$ and $(3, 17)$. Written as $y = mx + b$, the value of $m \times b$ is _____.

14. Determine the equation, in the form $y = mx + b$, of the line through $(-8, -19)$ and $(4, 17)$.

Answer Key. 1. (a) $m = 3$ (b) $m = -2$ (c) $m = 2$ (d) $m = -2$. **2.** (a) $y = 3x + 1$ (b) $y = 4x - 7$ (c) $y = 3x - 2$ (d) $y = -3x + 3$ (e) $y = 2x - 2$ (f) $y = -2x + 1$. **3.** (a) $y = 4x - 5$ (b) $y = -3x + 1$ (c) $y = 2x + 8$. **4.** (a) $y = \frac{1}{2}x + 4$ (b) $y = -\frac{2}{3}x + 7$ (c) $y = \frac{6}{5}x - 3$ (d) $y = -\frac{1}{2}x + 3$. **5.** (a) $y = 2x - 6$ (b) $y = -2x + 9$. **6.** (a) $y = 5$ (b) The run $x_2 - x_1 = 7 - 7 = 0$ and division by 0 is undefined, so the slope formula fails; equal x-coordinates give a *vertical* line, which has no $y = mx + b$ form — its equation is $x = 7$. **7.** (a) $d = 6t + 3$ (b) slope 6: she rides 6 km farther each hour (her speed); d-intercept 3: she was already 3 km from home when timing began (c) 51 km. **8.** (a) $p = 6t + 36$ (b) 96%. **9.** $h = 20a$. **10.** A. **11.** A. **12.** 11. **13.** 20. **14.** $y = 3x + 5$.

Lesson 7 Assignment *Show your work.*

Part A — Converting Between the Two Forms

1. Convert each equation from slope-intercept form ($y = mx + b$) to general form ($Ax + By + C = 0$), where A, B, and C are integers and A is positive.

(a) $y = 3x - 4$

(b) $y = -5x + 2$

(c) $y = x - 6$

(d) $y = -2x - 7$

2. Convert each equation from slope-intercept form to general form. Clear fractions first.

(a) $y = \frac{1}{4}x + 5$

(b) $y = -\frac{2}{5}x + 3$

(c) $y = \frac{3}{2}x - \frac{1}{4}$

3. Convert each equation from general form to slope-intercept form.

(a) $x - y - 3 = 0$

(b) $2x - y + 5 = 0$

(c) $x + y - 8 = 0$

(d) $3x + y + 1 = 0$

4. Convert each equation from general form to slope-intercept form. Simplify the slope fully.

(a) $3x - 2y - 8 = 0$ (b) $5x + 4y - 8 = 0$ (c) $2x - 3y + 9 = 0$

Part B — Intercepts and Proper Form

5. Find the x-intercept and the y-intercept of each line directly from its general-form equation (do not convert to slope-intercept form first).

(a) $2x + y - 6 = 0$

(b) $3x - y - 9 = 0$

(c) $x + 4y - 8 = 0$

(d) $5x - 2y - 10 = 0$

6. Each equation below is written with a negative value of A. Rewrite each one in proper general form so that A is positive.

(a) $-4x + y - 9 = 0$

(b) $-x - 3y + 12 = 0$

Part C — Mixed Review

7. A line has equation $5x + 2y - 10 = 0$. State its slope and its y-intercept.

8. A line has equation $6x - 3y - 18 = 0$. Without converting to slope-intercept form, state its x-intercept and its y-intercept.

9. Convert $y = -\frac{2}{3}x - 4$ to general form.

10. Convert $7x - 2y + 14 = 0$ to slope-intercept form.

Part D — Multiple Choice and Numerical Response

11. *(MC)* Which equation is written in proper general form, with A, B, C integers and A positive?
A. $4x + 2.5y - 3 = 0$ B. $y = 3x - 6$
C. $2x - 5y + 9 = 0$ D. $-3x + y - 4 = 0$

12. *(MC)* The slope of the line $8x - 4y + 12 = 0$ is
A. 2 B. -2 C. 3 D. -3

13. *(MC)* The y-intercept of the line $3x + 5y - 20 = 0$ is
A. -4 B. $\frac{20}{3}$ C. $-\frac{20}{3}$ D. 4

14. *(MC)* The x-intercept of the line $2x - 3y - 12 = 0$ is
A. 4 B. 6 C. -6 D. -4

15. *(NR)* A line has equation $kx + 4y - 16 = 0$ and passes through the point $(4, 0)$. The value of k is _____.

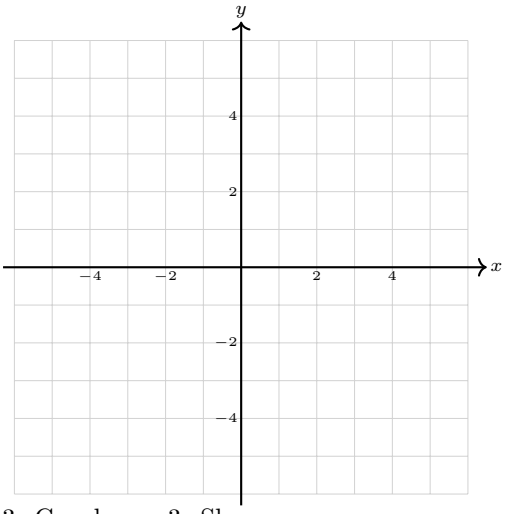
Answer Key. **1.** (a) $3x - y - 4 = 0$ (b) $5x + y - 2 = 0$ (c) $x - y - 6 = 0$ (d) $2x + y + 7 = 0$. **2.** (a) $x - 4y + 20 = 0$ (b) $2x + 5y - 15 = 0$ (c) $6x - 4y - 1 = 0$. **3.** (a) $y = x - 3$ (b) $y = 2x + 5$ (c) $y = -x + 8$ (d) $y = -3x - 1$. **4.** (a) $y = \frac{3}{2}x - 4$ (b) $y = -\frac{5}{4}x + 2$ (c) $y = \frac{2}{3}x + 3$. **5.** (a) x-int $(3, 0)$, y-int $(0, 6)$ (b) $(3, 0)$, $(0, -9)$ (c) $(8, 0)$, $(0, 2)$ (d) $(2, 0)$, $(0, -5)$. **6.** (a) $4x - y + 9 = 0$ (b) $x + 3y - 12 = 0$. **7.** slope $= -\frac{5}{2}$, y-int $= 5$. **8.** x-int $(3, 0)$, y-int $(0, -6)$. **9.** $2x + 3y + 12 = 0$. **10.** $y = \frac{7}{2}x + 7$. **11.** C. **12.** A. **13.** D. **14.** B. **15.** $k = 4$.

Lesson 8 Assignment *Show your work.*

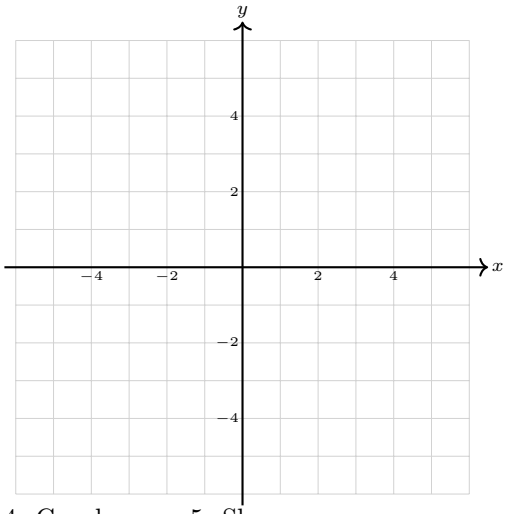
Part A — Horizontal and Vertical Lines

Graph each line in questions 1–4, and state the slope of the line.

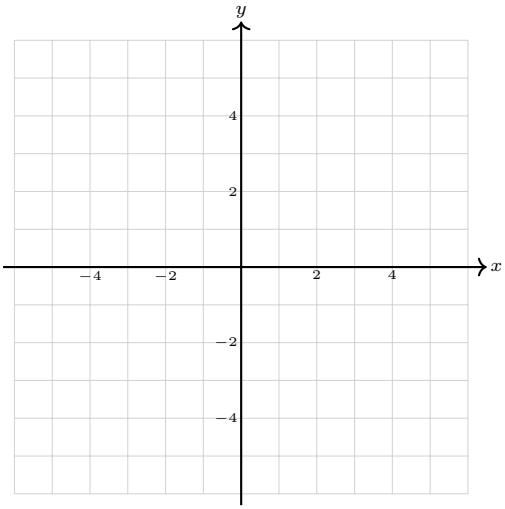
1. Graph $y = 6$. Slope: _____



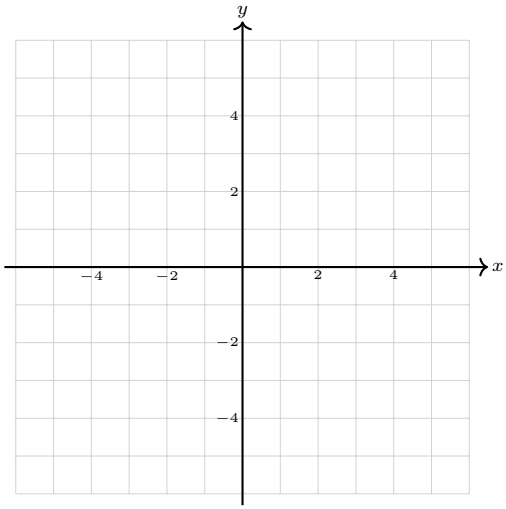
2. Graph $y = -5$. Slope: _____



3. Graph $x = 3$. Slope: _____

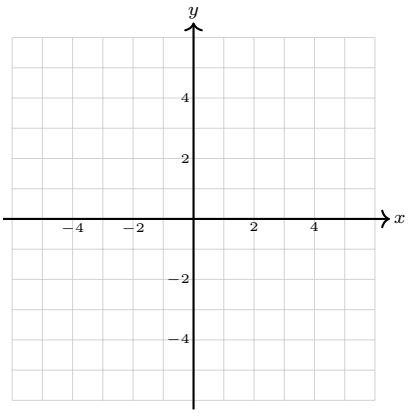


4. Graph $x = -5$. Slope: _____

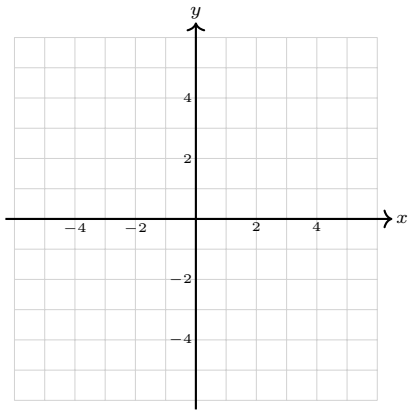


5. Graph each line below.

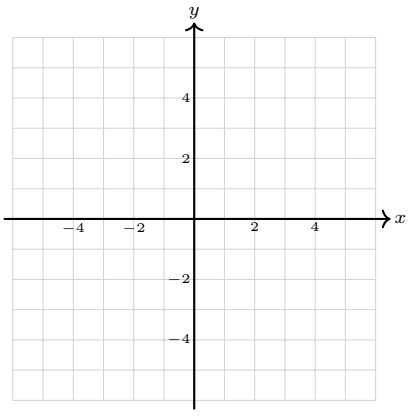
(a) $y = -2$



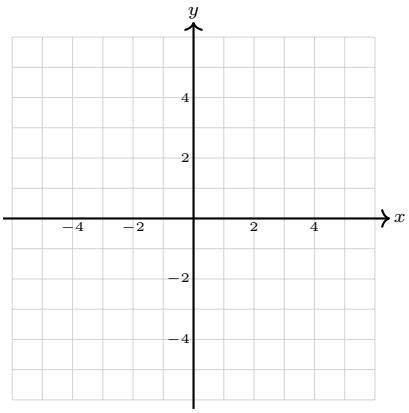
(b) $x = 4$



(c) $y = 0$



(d) $x = -1$

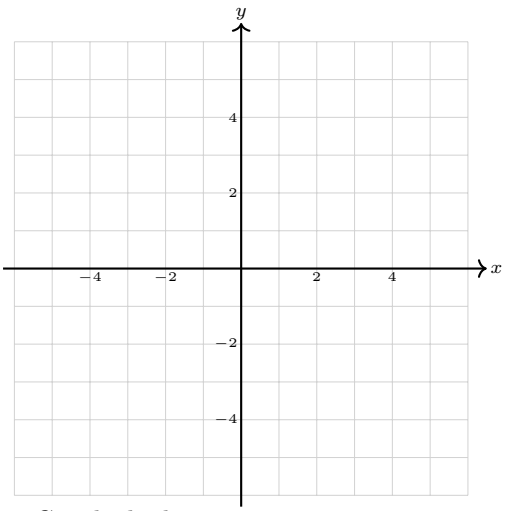


6. For each equation, state whether its graph is horizontal, vertical, or neither. If it is horizontal or vertical, give the coordinates of one point that must lie on it.

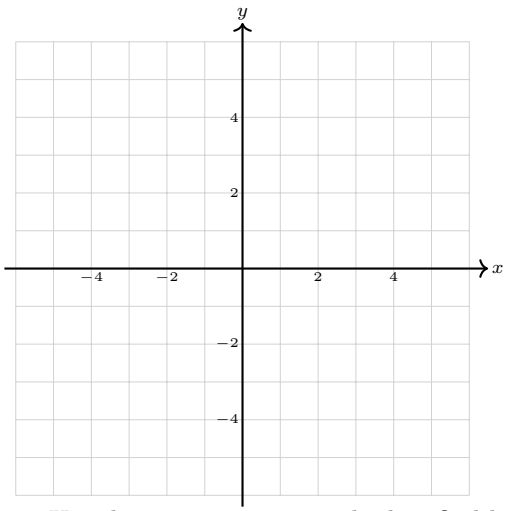
Equation	**Horizontal, Vertical, or Neither?**	**One Point on the Line**
(a) $x = -8$		
(b) $y = 3x + 2$		
(c) $y = 7$		
(d) $y = 0$		

Part B — Choosing and Applying a Strategy

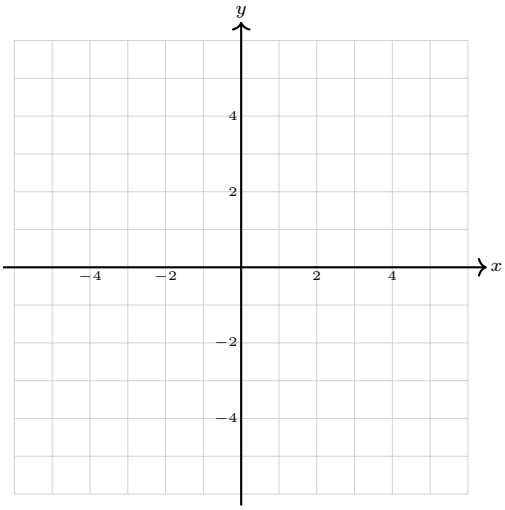
7. Explain which strategy you would use to graph $y = -3x + 5$ without altering its form, then graph it.
Strategy: _____



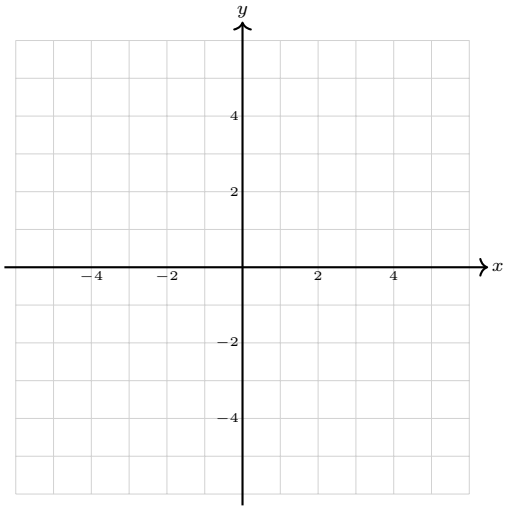
8. Explain which strategy you would use to graph $2x - y - 6 = 0$ (try the two-intercept method), then graph it.
x-int: _____ y-int: _____



9. Graph the line $x = -2$.
Slope: _____

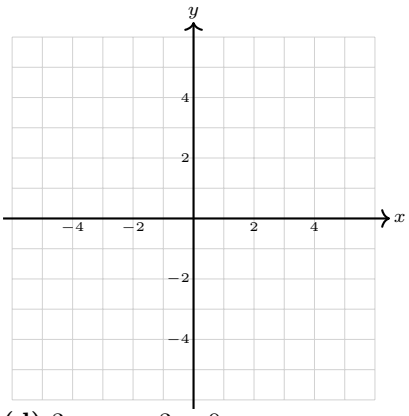


10. Use the two-intercept method to find both intercepts of $3x + 6y - 12 = 0$, then graph the line.
x-int: _____ y-int: _____

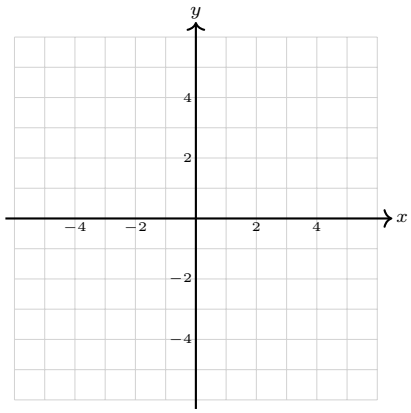


11. Graph each equation below, choosing whichever strategy is most efficient for its form.

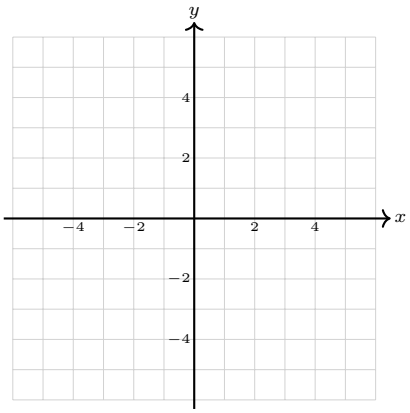
(a) $y = 2x - 4$



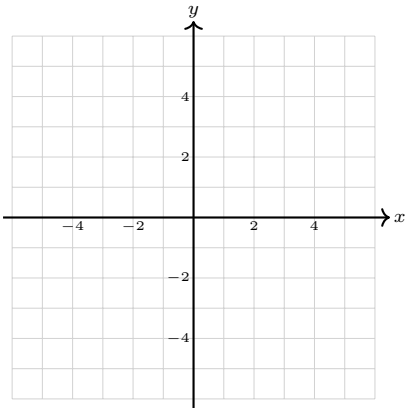
(b) $y = -1$



(c) $x = 5$



(d) $3x - y - 3 = 0$



Part C — Multiple Choice and Numerical Response

12. *(MC)* Which equation graphs as a vertical line?
A. $y = 5$ B. $x = -2$ C. $y = 2x$ D. $y = x - 2$

13. *(MC)* Which equation represents the *horizontal* line passing through $(3, -7)$?
A. $x = 3$ B. $y = -7$ C. $y = 3$ D. $x = -7$

14. *(MC)* A line has general form $2x - y + 6 = 0$. Its x-intercept is
A. -3 B. 3 C. -6 D. 6

15. *(NR)* The vertical line passing through $(4, -9)$ has equation $x = k$. The value of k is _____.

16. *(NR)* The equation $4x - 2y + 8 = 0$ is rewritten in slope-intercept form $y = mx + b$. The value of $m + b$ is _____.

Answer Key. 1. Horizontal line through $(0, 6)$; slope 0. **2.** Horizontal through $(0, -5)$; slope 0. **3.** Vertical through $(3, 0)$; slope undefined. **4.** Vertical through $(-5, 0)$; slope undefined. **5.** (a) horizontal at $y = -2$ (b) vertical at $x = 4$ (c) horizontal at $y = 0$ (the x-axis) (d) vertical at $x = -1$. **6.** (a) vertical, e.g. $(-8, 0)$ (b) neither (c) horizontal, e.g. $(0, 7)$ (d) horizontal (the x-axis), e.g. $(5, 0)$. **7.** Slope-intercept: plot $(0, 5)$, then count slope -3 (down 3, right 1). **8.** Two-intercept: x-int 3, y-int -6; join $(3, 0)$ and $(0, -6)$ (same line as $y = 2x - 6$). **9.** Vertical line through $(-2, 0)$; slope undefined. **10.** x-int 4, y-int 2; join $(4, 0)$ and $(0, 2)$. **11.** (a) through $(0, -4)$, slope 2 (b) horizontal at $y = -1$ (c) vertical at $x = 5$ (d) through $(0, -3)$, slope 3. **12.** B. **13.** B. **14.** A. **15.** 4. **16.** 6 (since $y = 2x + 4$).

Lesson 9 Assignment *Show your work.*

Part A — Interpreting Slope and y-Intercept

Answer each in a full sentence that names the quantity, the rate or value, and the units.

1. A cyclist's distance is $d = 18t$, where d is in km and t is in hours. State the slope and explain what it means, including units.

2. A leaking bucket holds $W = -50t + 800$, where W is in mL and t is in minutes. State the slope and explain what it means, including units. Is the water increasing or decreasing?

3. A candle's height is $H = -0.3t + 18$, where H is in cm and t is minutes burning. What is the y-intercept, and what does it represent?

4. A savings balance is $S = 25w + 200$, where S is in dollars and w is the number of weeks. What is the y-intercept, and what does it represent?

Part B — Writing and Using Equations

5. A dog walker charges a \$10 flat fee plus \$4 per dog walked.
(a) Write an equation for the total charge C in terms of the number of dogs walked, n.
(b) Determine the charge for walking 6 dogs.
(c) If a customer paid \$42, how many dogs were walked?

6. A moving company charges a \$75 flat fee plus \$40 per hour.
(a) Write an equation for the total cost C in terms of the number of hours, h.
(b) Find the cost of a 3-hour move.
(c) If a move cost \$315, how many hours did it take?

7. A water tank starts with 900 L of water and drains at a constant rate of 30 L per minute.
(a) Write an equation for the volume V remaining after t minutes.
(b) How much water remains after 12 minutes?
(c) How long until the tank is empty?

8. A phone plan costs \$20 per month plus \$0.05 per text message sent.
(a) Write an equation for the monthly cost C in terms of the number of texts, n.
(b) Find the cost in a month with 150 texts sent.
(c) One month the bill was \$31.50. How many texts were sent?

Part C — Comparing Two Situations

9. Two ride-share companies charge as follows: Company A charges \$3 plus \$1.50 per km; Company B charges \$6 plus \$1.00 per km.

(a) Write an equation for each company's cost.

(b) Determine the cost of a 10 km trip with each company.

(c) Find the distance at which both companies charge the same amount.

(d) Which company is cheaper for a trip shorter than the break-even distance? Explain.

10. Gym A charges a \$50 sign-up fee plus \$25 per month. Gym B charges no sign-up fee, but \$35 per month.

(a) Write an equation for the total cost at each gym after t months.

(b) Find the break-even number of months.

(c) For 3 months, which gym is cheaper? For a full year (12 months)?

Part D — Multiple Choice and Numerical Response

11. *(MC)* A plumber charges $C = 45h + 60$. What does the value 60 represent?
A. the hourly rate
B. the total cost for one hour of work
C. a flat fee charged even for 0 hours of work
D. the number of hours worked

12. *(MC)* Which equation best models a parking lot charging \$5 just to enter, plus \$3 for each hour parked?
A. $C = 5h + 3$ B. $C = 3h + 5$
C. $C = 8h$ D. $C = 3h - 5$

13. *(MC)* A graph shows temperature (°C) on the vertical axis and time (hours) on the horizontal axis. The slope of the line represents a rate measured in
A. hours per °C B. °C
C. °C per hour D. hours

14. *(NR)* Renting a bike costs $C = 4t + 8$, with C in dollars and t in hours. The cost of renting for 5 hours, in dollars, is _____.

15. *(NR)* A grocery delivery costs $C = 2.50n + 6$, where n is the number of items. If a delivery cost \$26, the number of items delivered was _____.

16. *(NR)* Two candles have heights $H_1 = -0.5t + 20$ and $H_2 = -0.2t + 11$, with t in minutes. The time t, in minutes, at which both candles are the same height is _____.

Answer Key. 1. $m = 18$; distance increases at 18 km/h. **2.** $m = -50$; water decreases at 50 mL/min. **3.** $b = 18$ cm; the candle's height before it was lit. **4.** $b = \$200$; the starting balance, before any weeks pass. **5.** (a) $C = 4n + 10$ (b) \$34 (c) 8 dogs. **6.** (a) $C = 40h + 75$ (b) \$195 (c) 6 h. **7.** (a) $V = -30t + 900$ (b) 540 L (c) 30 min. **8.** (a) $C = 0.05n + 20$ (b) \$27.50 (c) 230 texts. **9.** (a) A: $C = 1.5d + 3$; B: $C = 1.0d + 6$ (b) A: \$18, B: \$16 (c) 6 km (d) A — it has the smaller flat fee, so it is cheaper under 6 km. **10.** (a) A: $C = 25t + 50$; B: $C = 35t$ (b) 5 months (c) B cheaper at 3 months (\$105 vs. \$125); A cheaper at 12 months (\$350 vs. \$420). **11.** C. **12.** B. **13.** C. **14.** 28. **15.** 8. **16.** 30.

Lesson 10 Assignment *Show your work.*

Part A — Tables, Slope-Intercept Basics, and Graphing

1. Complete a table of values for $y = -2x + 5$ using $x = -2, -1, 0, 1, 2$, and list the five resulting ordered pairs.

x	-2	-1	0	1	2
y					

Ordered pairs:

2. Does the point $(5, -2)$ lie on the line $y = -x + 3$? Justify with a substitution.

3. State the slope and y-intercept of $y = 7x - 9$.

4. State the slope and y-intercept of $y = -x + 12$.

5. Rearrange $6x + y = 5$ into slope-intercept form, then state the slope and y-intercept.

6. For $y = 3x - 6$, state the y-intercept as a point, then use the slope to state the coordinates of a second point on the line.

7. For $y = -\frac{1}{4}x + 2$, state the y-intercept as a point, then use the slope to state a second point on the line (moving right).

Part B — Writing Equations of Lines

8–13. Determine the equation of each line in the form $y = mx + b$.

8. slope 8, y-intercept -5

9. slope $-\frac{2}{3}$, y-intercept 9

10. slope 5 through the point $(2, 4)$

11. slope -3 through the point $(-2, 7)$

12. through $(1, 2)$ and $(4, 14)$

13. through $(-2, 11)$ and $(4, -1)$

Part C — General Form, Horizontal and Vertical Lines

14. Convert $y = 6x - 11$ to general form.

15. Convert $5x - 3y - 15 = 0$ to slope-intercept form, simplifying the slope fully.

16. Determine the x- and y-intercepts of $4x + 5y - 20 = 0$ directly, without converting first.

17. Is the graph of $y = -8$ horizontal or vertical? Give one point on it.

18. Is the graph of $x = 9$ horizontal or vertical? Give one point on it.

19. For $3x - y - 9 = 0$, state the x- and y-intercepts you would plot to graph it by the two-intercept method.

Part D — Slope in Context

20. A cell-phone plan costs \$30 to start, plus \$0.05 per minute of calls. Let C be the total cost (dollars) after m minutes. Write an equation for C in terms of m, then state what the slope and the C-intercept represent.

21. A tank contains 500 L of water and is being drained at a constant rate. After 4 minutes it contains 380 L. Determine the drain rate (the slope) in L/min.

Part E — Mixed Review

22. A hot tub already contains 200 L of water and fills at a constant 45 L per minute.
(a) Write an equation for the volume V (litres) after t minutes.

(b) Determine the volume after 6 minutes.

(c) How many minutes until the hot tub reaches 605 L?

23. A candle is 24 cm tall when lit and burns at a constant 3 cm per hour. Let h be its height (cm) after t hours.
(a) Write an equation for h in terms of t.

(b) Determine the candle's height after 5 hours.

(c) When will the candle burn out completely (when will $h = 0$)?

24. A tree measured 190 cm tall 2 years after planting and 310 cm tall 5 years after planting, growing at a constant rate.
(a) Determine an equation for the height H (cm) in terms of the years y since planting.

(b) State what the slope represents in this situation.

(c) Predict the height of the tree when it is 9 years old.

25. Segment CD joins $C(-2, -7)$ and $D(2, 5)$.
(a) Determine the slope of CD.

(b) Line L_1 is parallel to CD through $(1, -2)$. Determine its equation in slope-intercept form.

(c) Line L_2 is perpendicular to CD through $(0, 4)$. Determine its equation in general form.

26. The line through $(-5, 2)$ and $(1, -10)$ crosses the x-axis at point A and the y-axis at point B.
(a) Determine the equation of the line in general form.
(b) State the coordinates of A and B.

Answer Key. **1.** $(-2, 9), (-1, 7), (0, 5), (1, 3), (2, 1)$. **2.** Yes: $-(5) + 3 = -2$. **3.** $m = 7$, $b = -9$. **4.** $m = -1$, $b = 12$. **5.** $y = -6x + 5$; $m = -6$, $b = 5$. **6.** $(0, -6)$; e.g. $(1, -3)$. **7.** $(0, 2)$; e.g. $(4, 1)$. **8.** $y = 8x - 5$. **9.** $y = -\frac{2}{3}x + 9$. **10.** $y = 5x - 6$. **11.** $y = -3x + 1$. **12.** $y = 4x - 2$. **13.** $y = -2x + 7$. **14.** $6x - y - 11 = 0$. **15.** $y = \frac{5}{3}x - 5$. **16.** $(5, 0)$ and $(0, 4)$. **17.** Horizontal; e.g. $(0, -8)$. **18.** Vertical; e.g. $(9, 0)$. **19.** $(3, 0)$ and $(0, -9)$. **20.** $C = 0.05m + 30$; slope = \$0.05 per minute (call rate), C-intercept = \$30 (base cost). **21.** -30 L/min. **22.** (a) $V = 45t + 200$ (b) 470 L (c) 9 min. **23.** (a) $h = -3t + 24$ (b) 9 cm (c) after 8 h. **24.** (a) $H = 40y + 110$ (b) grows 40 cm per year (c) 470 cm. **25.** (a) 3 (b) $y = 3x - 5$ (c) $x + 3y - 12 = 0$. **26.** (a) $2x + y + 8 = 0$ (b) $A(-4, 0)$, $B(0, -8)$.